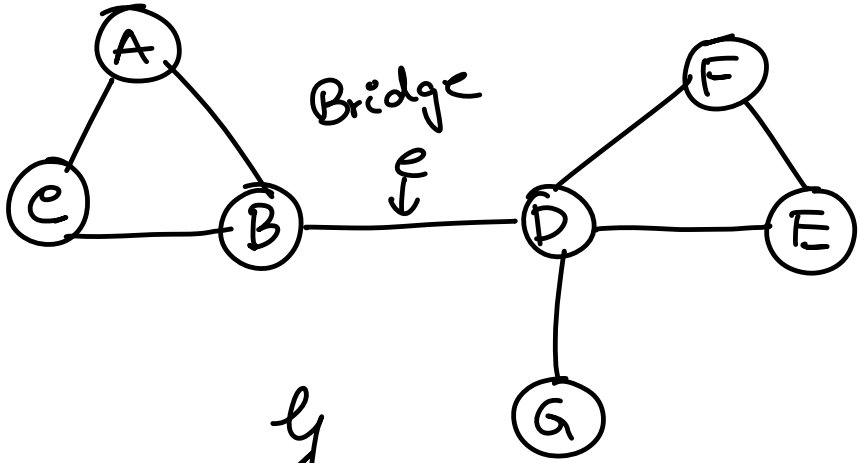


G

Bridges(G) = $\{BD, DG\}$.

CutPoints(G) = $\{B, D\}$

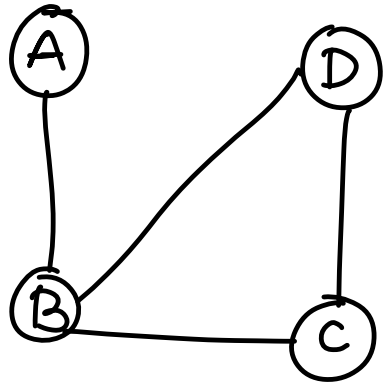
Bridge: An edge e is a bridge if deleting e increases the number of components

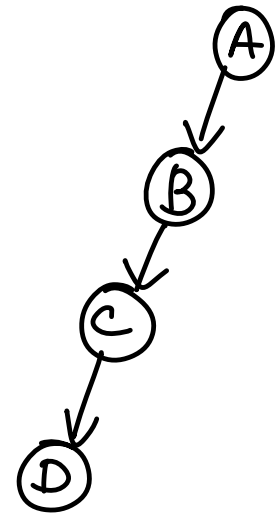
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Cutpoint (Articulation Point):

A vertex v is a cut point if deleting v and all edges incident with v increases the number of components.

Back edge:





DFS (Lexicographic)

AB, BC, CD

(DB). D → B exists in the graph, called back edge.

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Defn: An edge connecting a vertex to one of its ancestors in the DFS tree is called back edge

$\pi[u]$: Parent of u in the DFS tree.

$d[u]$: discovery time of u in DFS.

$low[u]$: back-reach of u

= The earliest discovery number reachable from the DFS subtree rooted at u.

S : Stack used

Tarjan Algorithm of Bridges and Cut Points:

DFS(u, Parent)

1. Mark u visited; assign d[u], low[u] ← d[u], push u onto S.

2. For each v ∈ Adj[u]

   If v = parent then

   continue // parent edge is ignored

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else if v is unvisited then // Tree edge

DFS(v, u)

low[u] ← min {low[u], low[v]}

if low[v] > d[u] then

uv is a bridge

if u is not a root and

low[v] >= d[u] then

u is a cut point

else if v is in S then // back edge

$$low[u] \leftarrow \min\{low[u], d[v]\}$$

3. Pop u from S

4. Root is a cutpoint iff it has atleast 2 DFS children.

Formulae:

Back edge: uv

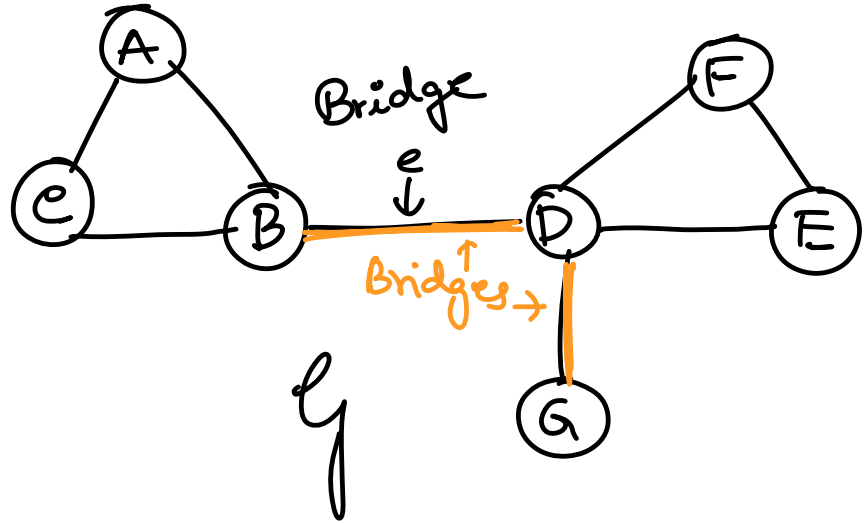
$$low[u] \leftarrow \min\{low[u], d[v]\}$$

Return from child v:

$$low[u] \leftarrow \min\{low[u], low[v]\}$$

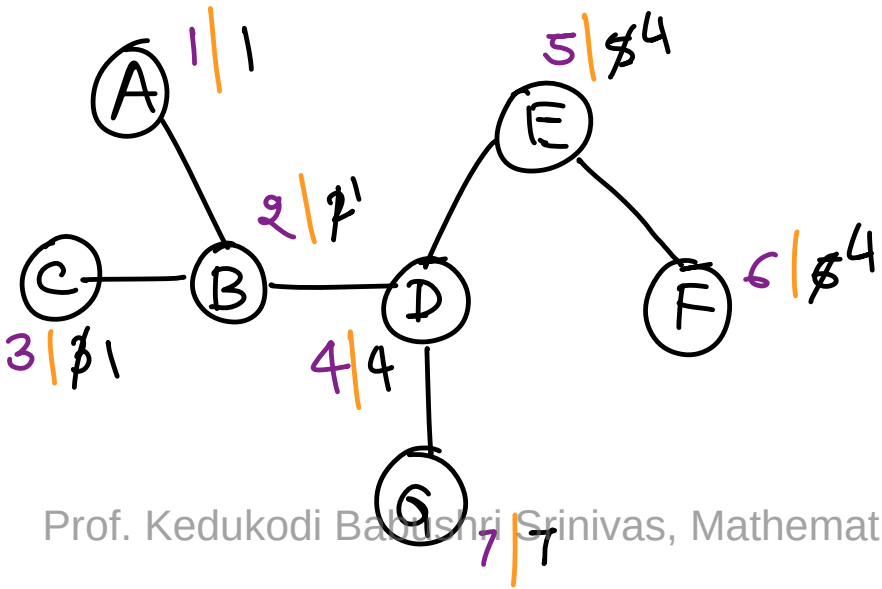
Stack S is written in bottom $\rightarrow$ top order

bottom, ..., top.



G

DFS TREE:



Visited order by DFS | low [v]
d[v]

Lexicog. order followed.

A → B → C
↓
D → E → F
↓
G

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Steps	Action	Stack S Status	low-link computation
1.	Discover u=A	u = A	$d[A] = 1 = low[A]$
2.	Discover B	A, B	$d[B] = 2 = low[B]$
3.	Discover C	A, B, C	$d[C] = 3 = low[C]$
4.	Test A $A \in S$ $\Rightarrow$ CA is a back edge	A, B, C	backedge $uv=CA$ $low[C] = \min\{low[u], d[v]\}$ $= \min\{low[C], d[A]\}$ $= \min\{3, 1\}$ $= 1$
5.	Pop C	A, B	Return from child C $low[B] = \min\{low[u], low[v]\}$ $= \min\{low[B], low[C]\}$ $= \min\{2, 1\} = 1.$

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6.	Discover D	A, B, D	low[D] = d[D] = 4.
7.	Discover E	A, B, D, E	low[E] = d[E] = 5
8.	Discover F	A, B, D, E, F	low[F] = d[F] = 6
9.	Test D D ∈ S ⇒ FD is a back edge	A, B, D, E, F	back edge uv = FD low[F] = min{low[F], d[D]} = min{6, 4} = 4
10.	Pop F	A, B, D, E	Return from child F low[E] = min{low[F], low[E]} = min{4, 5} = 4

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11.	Pop E	A, B, D	Return from child E $low[D] = \min\{low[E], low[D]\}$ $= \min\{4, 4\}$ $= 4$
12.	Discover G	A, B, D, G	$d[G] = low[G] = 7$
13.	Pop G	A, B, D	$low[D] \leftarrow \min\{\overset{low[D]}{4}, \overset{low[G]}{7}\}$ $= 4$
14.	Pop D	A, B	$low[B] = \min\{1, 4\}$ $= 1$
15.	Pop B	A	$low[A] = \min\{1, 1\} = 1.$

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16.	Pop A	ϕ	Done

Bridges: $u \to v$

$low[v] > d[u]$

Edges in DFS	Test $low[v] > d[u]$	Result
AB	$low[B] = 1 \not> 1 = d[A]$	No
BC	$low[C] = 1 \not> 2 = d[B]$	No
✓ BD	$low[D] = 4 > 2 = d[B]$	Bridge
DE	$low[E] = 4 \not> 4 = d[D]$	No
EF	$low[F] = 4 \not> 5 = d[E]$	No

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✓ DG	low[G] = 7 > 4 = d[D]	Bridge

Bridges = $\{BD, DG\}$.

Cut Points:

u	Child V	Test $low[v] \geq d[u]$	Result
A	B	Root ≥ 2 children	No
B	C	$1 \not\geq 2$	No
B	D	$B \to D$ $low[D] = 4 \geq 2 = d[B]$	B is cut point
D	E	$4 \geq 4$	D is cutpoint
D	G	$7 \geq 4$	D is cutpoint
E	F	$4 > 5$	No

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